

# Electromagnetic spectrum and space

## *Teaching guide*

### Learning objectives

- The student will be able to describe the purpose of using radio telescopes
- the student will be able to define radiation
- The student will be able to define the electromagnetic spectrum?

### Materials

- Colors we See activity (spectrum, stickers?)
- EM Spectrum Worksheet
- Colored Pencils/crayons- Red, Blue, Yellow, Green, Purple
- Manipulative Terms

### 1. "Colors" we see Activity

- Background
- How do we "see" colors?
  - 1. Reflected vs. Absorbed
- Different Colors have Different properties
  - 1. Mix Red and Yellow= Orange
  - 2. Mix Yellow and Blue = Green
- Property=wavelength
- Procedure:
- Head to head competition
- Students race to place sticker on portion of spectrum that we "see"
  - 1. Green Absorbed-students place stickers on green area of spectrum
  - 2. Yellow, Blue, Green-Reflected-students would place stickers on red, orange and purple.
  - 3. All colors reflected- white
  - 4. All colors absorbed-black

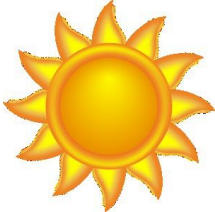
### Transition: small to large scale

### 2. Studying Space

- Space Studied?
- Telescopes
- Special Telescope- Radio Telescope

### 3. What is being "viewed/studied" for space?

- Sun
- Heat
  - 1. Heat=tells us Energy is present
- Energy-Radiation
- Cartoon Drawing of Sun



### 4. Organizing Space Information

- Electromagnetic Spectrum
- Electro-Electricity
- Spectrum-arranges it
- Magnetic-magnetism
- Everything in space= gives off Radiation (ENERGY)

### 5. Electromagnetic Spectrum

- Manipulative
- Students color visible light portion